

EGG EXTRUSION, EMBRYO DEVELOPMENT, TIMING AND DURATION OF ECLOSION, AND INCUBATION PERIOD OF PRIMIPAROUS AND MULTIPAROUS TANNER CRABS (*CHIONOECETES BAIRDI*)

Katherine M. Swiney

National Marine Fisheries Service, Alaska Fisheries Science Center, Kodiak Fisheries Research Center, Kodiak Laboratory,
301 Research Court, Kodiak, Alaska 99615, U.S.A. (katherine.swiney@noaa.gov)

ABSTRACT

This study compares timing of egg extrusion, embryo development, timing and duration of eclosion, and incubation periods of Kodiak, Alaska, primiparous and multiparous Tanner crabs (*Chionoecetes bairdi*) reared in identical conditions to determine if and how these variables differ between reproductive states. Female reproductive state (primiparous or multiparous) and dates of egg extrusion were recorded, eggs were sampled monthly to determine egg stage and area, and larvae were collected daily during eclosion to determine timing and duration of eclosion. Primiparous Tanner crabs extruded eggs on average 103 days earlier than multiparous females and embryos developed similarly between groups. Both groups exhibited an embryonic developmental diapause at the gastrula stage, but the length of diapause was approximately 6 months for the primiparous group and 3 months for the multiparous group. This diapause appears to synchronize eclosion. The eggs of primiparous Tanner crabs were significantly smaller than the eggs of multiparous females for a few months after extrusion, but the differences may not be biologically significant. Eclosion was relatively synchronous between the two groups, however the mean eclosion date was 10 days earlier for primiparous females and on average, eclosion took 7 days for primiparous females and 12 days for multiparous females. Primiparous Tanner crabs have an average incubation period of 489 days which is 92 days longer than the average multiparous female incubation period of 397 days. Results of this study clearly illustrate that reproductive cycles differ between primiparous and multiparous Tanner crabs which may result in different reproductive potentials. Differences between primiparous and multiparous Tanner crabs must be understood and included in models for effective stock assessment, fishery management plans, and rebuilding plans.

KEY WORDS: *Chionoecetes bairdi*, diapause, eclosion, egg extrusion, embryo development, incubation period, primiparous, multiparous, Tanner crabs

INTRODUCTION

Tanner crabs, *Chionoecetes bairdi* Rathbun, 1924, have historically supported valuable commercial fisheries in Alaskan waters, but because the stock in the eastern Bering Sea is considered overfished (Rugolo et al., 2006) and there is a reduced stock size around Kodiak Island (Sagalkin, 2004) these fisheries were closed. The eastern Bering Sea fishery, closed in 1997 (Alaska Department of Fish and Game, 2004), is considered overfished and is currently under the rebuilding plan approved by the Alaska Board of Fisheries and the North Pacific Fishery Management Council for the Bering Sea Tanner crab stock (Rugolo et al., 2006). In the eastern Bering Sea, a small fishery re-opened in the Western District in 2005 and small fisheries in the Eastern and Western Districts were opened in 2006 (Rugolo et al., 2006). The fishery around Kodiak Island closed in 1994 and sections began to re-opened in 2001 (Sagalkin, 2004). The Kodiak area Tanner crab commercial harvest peaked in the 1977/78 season when over 15 million kg were harvested with an ex-vessel value of U.S. \$14.3 million (Sagalkin, 2004). In contrast, during the 2004/05 season, Kodiak area fishers harvested 818,522 kg of Tanner crab with an ex-vessel value of U.S. \$3.11 million (Sagalkin, 2006). Because some Alaskan Tanner crab stocks are considered overfished or at a reduced stock size, it is vital to understand fundamental biological parameters such as reproduction to rebuild stocks and effectively manage this species.

Female Tanner crabs have a terminal molt that is thought to occur at approximately age five (Donaldson, 1977; Donaldson and Adams, 1989). Fifty percent maturity for Kodiak area females is 83 mm carapace width (CW) (Donaldson, 1977). Tanner crabs have a premating embrace in which a mature male grasps a pubescent female, one that has not yet undergone the terminal molt (Donaldson and Adams, 1989). The male guards and aids the female while she molts then mating occurs shortly thereafter (Donaldson and Adams, 1989). Within 48 h of mating, a female extrudes a fertilized clutch of eggs that she broods externally until eclosion. A female brooding her first egg clutch is classified as primiparous and a multiparous female has brooded eggs at least once in previous years. Tanner crab females can store sperm for at least two years and mature females can either mate while hard shelled or use stored sperm to fertilize subsequent egg clutches (Paul, 1984; Paul and Adams, 1984; Paul et al., 1996). Fecundity estimates range from 24,000 to 318,000 eggs (Hilsinger, 1976) with primiparous Tanner crabs having a lower fecundity. Primiparous females produce 62% (Wenner and Dugan, 1991) to 70% (Somerton and Meyers, 1983) as many eggs as equal-sized multiparous females.

Egg extrusion among female Tanner crabs is not synchronous, but eclosion appears to be relatively synchronous in this species. Near Kodiak, Alaska the majority of pubescent females in mating pairs are found from mid-November through mid-July and all multiparous mating

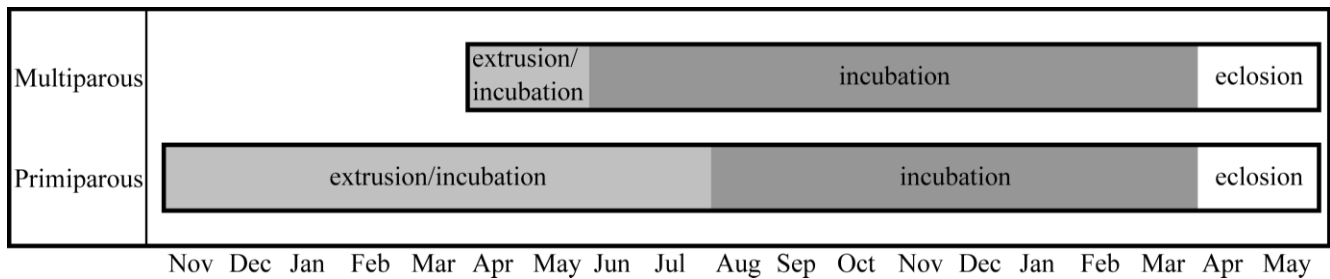


Fig. 1. Diagram of primiparous and multiparous Tanner crab reproductive cycle illustrating asynchronous egg extrusion, longer mating season and incubation period for primiparous females and relatively synchronous eclosion. "Extrusion/incubation" designates the months that extrusion occurs and therefore, incubation begins. For example, if a female extrudes in November her incubation period begins immediately after extrusion in November. "Incubation" is a continuation of the incubation period that began after extrusion and differs from "extrusion/incubation" in that females are no longer extruding eggs during these months. Extrusion and incubation period data based upon NMFS unpublished data. Eclosion data is from Hilsinger (1976), Donaldson (1977), Incze et al. (1982), and Donaldson and Adams (1989).

pairs are found from late April through May (NMFS unpublished data). Based upon these observations, pubescent females in mating pairs are expected to undergo a terminal molt, mate and extrude their first egg clutch, becoming primiparous from mid-November through mid-July. On the other hand, multiparous females are expected to extrude new egg clutches from April through May. Eclosion is generally agreed to occur the following year from April through May for both groups (Hilsinger, 1976; Donaldson, 1977; Incze et al., 1982; Donaldson and Adams, 1989) (Fig. 1).

Earlier egg extrusion of primiparous females (NMFS unpublished data) and relatively synchronous eclosion of primiparous and multiparous larvae (Hilsinger, 1976; Donaldson, 1977; Incze et al., 1982; Donaldson and Adams, 1989) results in primiparous females having a longer incubation period (Fig. 1). It is currently unknown what occurs during embryo development of primiparous and multiparous Tanner crabs; however it is likely that embryos of primiparous females either develop at a slower rate than those of multiparous females or the embryos of primiparous females go into a state of diapause during development.

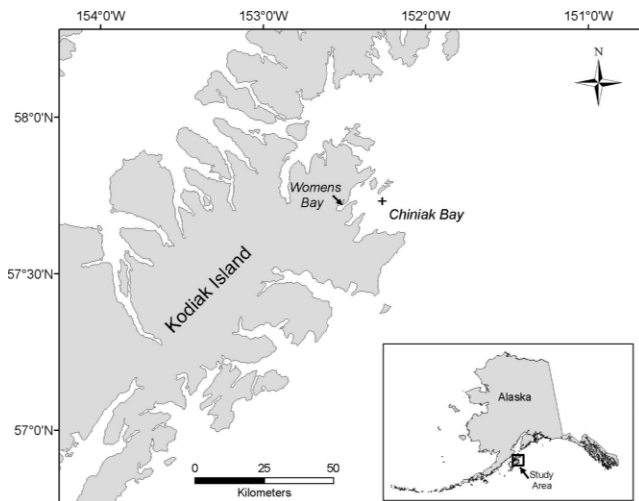


Fig. 2. Locations where Tanner crabs were collected. Primiparous females were collected from Womens Bay and multiparous females were from Chiniak Bay.

Crab reproductive studies typically do not differentiate between primiparous and multiparous crabs or utilize only multiparous females, but when examined, differences in reproductive cycles are found between primiparous and multiparous crabs (Sainte-Marie, 1993; Moriyasu and Lanteigne, 1998; Stevens and Swiney, 2007). An understanding of reproduction and its effects on parameters such as reproductive potential between reproductive states is important for effective stock assessment and fishery management plans, including stock rebuilding plans, for crab species. Differences between primiparous and multiparous Tanner crab reproductive cycles and eclosion is largely unknown. In this study, timing of egg extrusion, embryo development, timing and duration of eclosion, and incubation period were compared between captive primiparous and multiparous Tanner crabs exposed to identical conditions to determine if and how these variables differ between reproductive states.

MATERIALS AND METHODS

Sample Collection and Laboratory Study

Female Tanner crabs were captured in the field November 2002 through May 2003 and brought to the Kodiak Fisheries Research Center (KFRC) laboratory. Females were grouped as either primiparous or multiparous. Females that comprised the multiparous group were collected by commercial crab pots fished in Chiniak Bay, Kodiak, Alaska (57°43.25'N, 152°17.5'W, Fig. 2), November, 2002 and delivered to the laboratory. The females collected were ovigerous, hatched their larvae in the laboratory and extruded a new clutch of eggs. Once a female began stripping her pleopods after eclosion, she was paired with a mature male; therefore, the female could either mate or use stored sperm to fertilize a new clutch. It is not known how many previous clutches females in the multiparous group brooded, but they were multiparous because they were collected ovigerous and then extruded a new clutch of eggs in the laboratory. Primiparous females were collected by divers while still pubescent and nearing their terminal molt (as indicated by being in a grasping pair) in Womens Bay, Kodiak, Alaska (57°43.6'N, 152°32.0'W, Fig. 2). Divers searched for pubescent females in mating pairs from November 2002 through May 2003; however, females used in this study were collected between December 2002 and February 2003, except for one female collected in April 2003. The grasping pairs were isolated in underwater cages that were monitored regularly until a female underwent her terminal molt, mated and extruded eggs after which time she was brought into the laboratory. Occasionally, a female underwent her terminal molt in a cage without a male present. When this occurred, the soft shelled female was brought into the laboratory and mated with a male recently collected from the field.

All females were tagged with an oval 6.5 mm × 20 mm individually numbered plastic Floy tag [Reference to trade names does not imply endorsement by the National Marine Fisheries Service, NOAA] attached

Table 1. Sample sizes of primiparous and multiparous crabs by month and year.

	Primiparous	Multiparous
Jan 03	14	0
Feb 03	21	0
Mar 03	21	0
Apr 03	24	0
May 03	22	21
Jun 03	23	21
Jul 03	23	21
Aug 03	21	15
Sep 03	20	14
Oct 03	19	12
Nov 03	19	10
Dec 03	18	9
Jan 04	16	9
Feb 04	14	9
Mar 04	14	9
Apr 04	14	9

with a cable tie to a fourth walking leg. Data recorded were female reproductive state (primiparous or multiparous), molting and mating dates, dates of egg extrusion and carapace width measured as the greatest straight-line distance across the carapace excluding spines. Mean extrusion dates were calculated as the average date of extrusion within the primiparous and multiparous groups.

For the duration of the study, crabs were held in a single flow-through tank 0.46 m × 0.46 m × 1.8 m at the KFRC seawater laboratory. Seawater intakes are located at 15 and 26 m depths in Trident Basin approximately 30 m from the laboratory. Both primiparous and multiparous females were held in the same tank to ensure that all crabs in the experiment were exposed to identical conditions. Sand-filtered seawater was used and the tank was chilled from May 2003 through November 2003 to ensure appropriate temperatures. An Onset StowAway TidBit data logger recorded water temperature in the tank. Crabs were fed ad libitum twice weekly a diet of fish and squid.

Twenty-four primiparous females and 21 multiparous females were used in this experiment. However, crabs died over the course of the experiment, leaving 14 primiparous and 9 multiparous females at the end (Table 1). Crabs were sampled on the 15th (± 2) day of every month beginning in January 2003 and ending April 2004, when eclosion occurred. Upon visual examination, all crabs and their eggs appeared healthy when they were sampled. Females died over the course of the experiment, but there was no obvious reason to eliminate the eggs of these females (collected before death) from the study since the eggs appeared to be healthy and developing normally when collected.

Monthly sampling was a two-step process in which embryo developmental stages were determined and digital images of eggs were captured. A randomly sampled egg clump was removed from each female with forceps and approximately 20 eggs were immersed in Bouin's solution for 5 min to facilitate observation of the external morphology of the embryos (Moriyasu and Lanteigne, 1998). After embryos reached the eyed stage, they were not placed in Bouin's solution. Embryo developmental stages were determined using a compound microscope at a total magnification of 50× and developmental stages followed those of Moriyasu and Lanteigne (1998) for snow crabs, *Chionoecetes opilio* (Fabricius, 1788). Embryo developmental stages are provided in Table 2. Additionally, digital images of 10 fresh eggs from each female were taken with a digital camera attached to a compound microscope at a total magnification of 50×; only fresh eggs were used because preserved eggs swell (Moriyasu and Lanteigne, 1998). Image analysis software (Image Pro Plus Version 4.1 for Windows, Media Cybernetics, Inc. 8484 Georgia Avenue, Suite 200, Silver Spring, Maryland 20910) was used to measure egg area rather than diameter to reduce measurement bias due to the imperfect sphericity of the eggs.

At the end of April 2004, females were placed in individual containers with flow-through seawater at ambient temperature, and nets (101.5 mm with 350 micron mesh) were placed on the seawater outflow to retain all the larvae. Newly hatched larvae were collected daily from each female as in Stevens and Swiney (2007); the amount of larvae released was determined by measuring volume. Larvae from each net were transferred to a graduated cylinder and seawater was added; after the larvae settled, the volume (mL)

Table 2. Embryo developmental stages used in this study based upon Moriyasu and Lanteigne (1998) descriptions of embryo developmental stages of *Chionoecetes opilio*. See Moriyasu and Lanteigne (1998) for more detailed descriptions.

Stage #	Embryo developmental stage
1	Prefuniculus formation
2	Funiculus formation
3	Cleavage and blastula
4	Gastrula
5	Lateral ectodermal band
6	Prenauplius
7	Nauplius
8	Maxilliped formation
9	Metanauplius
10	Late metanauplius
11	Eye-pigment formation
12	Chromatophore formation
13	Reduced yolk
14	Prehatching

of larvae in the cylinder was recorded. For the purposes of analysis in this study, eclosion began when 0.1 mL of larvae were collected and ended when females began stripping their pleopods clean which coincided with the last of the larvae hatching. Mean hatch date was the weighted average of larval output over time calculated by multiplying the daily volume of hatched larvae by day-of-the-year, summing the products over time and dividing by the total volume of larvae released. Incubation period was the number of days between when a female extruded eggs and the last day of hatching.

Data Analysis

Mean egg area and embryo developmental stages were calculated monthly by pooling all of the data within each group: primiparous and multiparous females. Because the same females were sampled repeatedly throughout the experiment, each month was treated as a single experiment to eliminate independence artifacts. Comparison of embryo developmental stages and egg areas were conducted for the months of May 2003 through April 2004 and were not conducted before May 2003 because prior to that date only primiparous females were found brooding a new clutch of eggs. Mean egg areas were compared between primiparous and multiparous females monthly using two-sample *t* tests; critical values were adjusted using Sidak's multiplicative inequality because multiple *t* tests were conducted (Rohlf and Sokal, 1981). Mann-Whitney *U* tests were computed monthly to detect differences between mean embryo development stages of primiparous and multiparous females. Developmental degree-days were calculated as the sum of the average daily temperatures over the incubation period for each crab. Mean extrusion date, mean hatch date, mean incubation period, mean developmental degree-days and days spent releasing larvae for primiparous and multiparous females were compared with two-sample *t* tests. Means with standard errors (SE) are reported. Statistics were calculated in SYSTAT (2002) and by methods described by Sokal and Rohlf (1995) and Rohlf and Sokal (1981) statistical tables. Values were considered significant when *P* < 0.05 and highly significant when *P* < 0.01.

RESULTS

Mean water temperature in the tank during the course of the experiment was 5.9°C (SE = 0.06) (Figs. 3, 4). A daily mean low of 2.8°C was observed 18 January 2004 and a daily mean high of 9.1°C was observed 19 July 2003. January through April was on average 1.6°C (SE = 0.26) colder in 2004 than 2003.

Primiparous females extruded eggs from December through February, except for one female which extruded eggs in late May, yielding a mean extrusion date of 10 January 2003 (SE = 6.04), whereas multiparous females only extruded eggs in April and the first week of May resulting in a significantly later (103 d) mean extrusion date

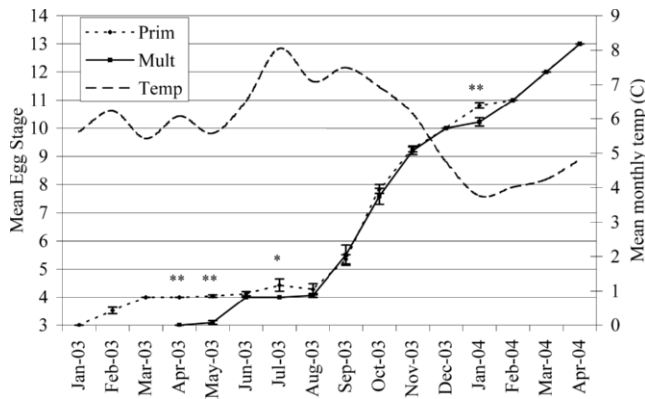


Fig. 3. Comparison of mean embryonic developmental stages between primiparous and multiparous Tanner crabs. Mean monthly temperatures are shown. "Prim" denotes primiparous females and "Mult" denotes multiparous females. See Table 2 for embryo developmental stages. Values are means and vertical bars represent standard errors. * $P < 0.05$; ** $P < 0.01$.

of 23 April 2003 ($SE = 1.35$) ($t = 4.2$, $df = 40.4$, $P < 0.01$, Table 3).

Embryos of primiparous and multiparous females developed similarly. Approximately 2 months after egg extrusion, the embryos of both groups developed to the gastrula stage (embryo developmental stage 4) at which point they went into a state of embryonic developmental diapause, a period of arrested development, through August 2003. Embryonic developmental diapause lasted approximately 6 months for the primiparous group and 3 months for the multiparous group (Fig. 3). Because primiparous females extruded eggs earlier than multiparous females, primiparous female embryonic development was at a significantly more advanced stage in April ($U_s = 0.0$, $P < 0.01$), May ($U_s = 21.0$, $P < 0.01$) and July 2003 ($U_s = 199.5$, $P < 0.05$, Fig. 3). After August 2003 the embryos of both groups developed rapidly and in synchrony except in January 2004 when the embryos from the primiparous group were significantly more developed ($U_s = 29.5$, $P < 0.01$, Fig. 3).

The eggs of primiparous females were significantly smaller than the eggs of multiparous females for May through August 2003 (May, $t = 6.2$, $df = 427.9$, $P < 0.01$; June, $t = 5.4$, $df = 437.8$, $P < 0.01$; July, $t = 3.8$, $df = 438.0$, $P < 0.01$; August, $t = 3.9$, $df = 263.8$, $P < 0.01$, Fig. 4). Beginning with the September 2003 sampling, there were no longer significant differences between the egg areas of the two experimental groups, except in April 2004 when the eggs of primiparous females were significantly larger than those of multiparous females ($t = 3.4$, $df = 225.7$, $P < 0.01$, Fig. 4). Although highly significant differences were detected in egg areas between the two groups May through August, the differences may not be biologically significant. The differences between the egg areas of both groups were only 0.009, 0.007, 0.006 and 0.008 mm May through August and it is difficult to discriminate the eggs of primiparous and multiparous females solely upon egg area.

Eclosion began on 28 April 2004 and ended on 7 June 2004 with mean eclosion dates for primiparous and multiparous females in May (Fig. 5). The 8 May 2004 ($SE = 1.25$) mean eclosion date observed among primiparous females was significantly earlier than the 18 May 2004

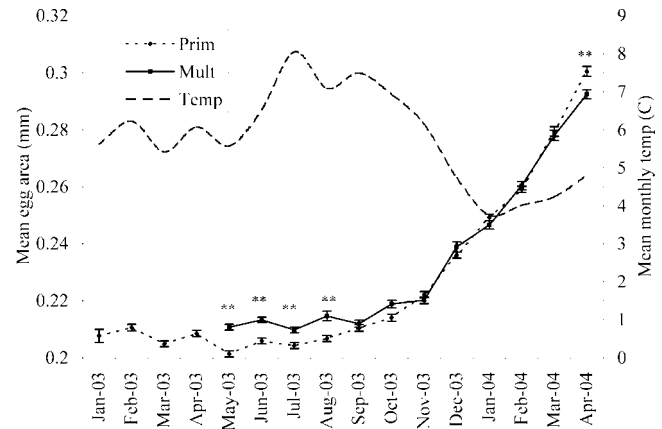


Fig. 4. Comparison of mean egg areas of primiparous and multiparous Tanner crabs. Mean monthly temperatures are shown. "Prim" denotes primiparous females and "Mult" denotes multiparous females. Values are means and vertical bars represent standard errors. ** $P < 0.01$.

($SE = 1.41$) mean eclosion date observed among multiparous females ($t = 4.9$, $df = 18.5$, $P < 0.01$, Table 3, Fig. 5). On average, eclosion among primiparous females required significantly less time, 7.2 days ($SE = 0.58$) in comparison to the 12 days of eclosion observed among multiparous females ($SE = 1.64$) ($t = 2.7$, $df = 10.0$, $P < 0.05$).

Incubation period and developmental degree-days differed between primiparous and multiparous female Tanner crabs primarily because of earlier egg extrusion by primiparous females and relatively synchronous eclosion. The incubation period of primiparous females was on average 489 ($SE = 4.93$) days, significantly longer than the average multiparous females incubation period of 397 ($SE = 1.46$) days, a difference of 93 days ($t = 18.0$, $df = 15.2$, $P < 0.01$, Table 3). The differences observed for the incubation periods of primiparous and multiparous Tanner crabs were also reflected in observed developmental degree-days. Primiparous females accumulated on average 2843.09 ($SE = 28.13$) developmental degree-days whereas multiparous females accumulated 2330.47 ($SE = 10.78$) developmental degree-days, a significant difference of 512.62 developmental degree-days ($t = 17.0$, $df = 16.5$, $P < 0.01$, Table 3).

DISCUSSION

Timing of egg extrusion, embryo development, timing and duration of eclosion, and incubation period differ between primiparous and multiparous Tanner crabs held in identical conditions. Primiparous females extrude eggs 103 days earlier in the year, have a three month longer embryonic diapause phase and a 93 day longer incubation period than multiparous females. Eclosion is relatively synchronous in this species, however the mean hatch date of primiparous females is 10 days earlier than for multiparous females and multiparous females require approximately five more days for eclosion.

In this study, multiparous females extruded eggs April through the beginning of May (Table 3) which is the time period consistently reported by other researchers (Hilsinger, 1976; Donaldson, 1977; Paul and Adams, 1984; Paul and Paul, 1992). The time period in which pubescent females

Table 3. Information on the timing of egg extrusion, mean eclosion dates, incubation periods and developmental degree days in primiparous and multiparous Tanner crabs. "Extrude" is the day eggs were extruded, "Eclosion" denotes the mean eclosion date, "Incubation" is the number of days eggs were incubated and "DD" is the developmental degree-days. Data is not available for some "Eclosion," "Incubation" and "DD" fields due to crabs dying before eclosion and is recorded as n/a. The dates or numbers in brackets denote mean values for the respective columns. Standard errors are reported.

Primiparous				Multiparous			
Extrude	Eclosion	Incubation	DD	Extrude	Eclosion	Incubation	DD
13 Dec 02	n/a	n/a	n/a	10 April 03	n/a	n/a	n/a
14 Dec 02	n/a	n/a	n/a	11 April 03	15 May 04	402	2359.13
17 Dec 02	12 May 04	514	2984.72	18 April 03	16 May 04	396	2323.64
19 Dec 02	11 May 04	510	2958.62	18 April 03	11 May 04	391	2289.10
25 Dec 02	01 May 04	495	2864.66	19 April 03	n/a	n/a	n/a
25 Dec 02	n/a	n/a	n/a	21 April 03	18 May 04	397	2331.92
26 Dec 02	06 May 04	498	2883.58	21 April 03	n/a	n/a	n/a
29 Dec 02	n/a	n/a	n/a	22 April 03	18 May 04	394	2311.26
31 Dec 02	01 May 04	489	2833.11	22 April 03	22 May 04	397	2332.55
31 Dec 02	14 May 04	501	2911.44	22 April 03	n/a	n/a	n/a
02 Jan 03	n/a	n/a	n/a	23 April 03	n/a	n/a	n/a
04 Jan 03	08 May 04	492	2856.68	25 April 03	n/a	n/a	n/a
07 Jan 03	n/a	n/a	n/a	27 April 03	n/a	n/a	n/a
07 Jan 03	12 May 04	492	2862.16	27 April 03	23 May 04	397	2337.89
08 Jan 03	10 May 04	489	2843.19	28 April 03	n/a	n/a	n/a
14 Jan 03	09 May 04	482	2807.85	28 April 03	n/a	n/a	n/a
14 Jan 03	09 May 04	483	2814.84	29 April 03	n/a	n/a	n/a
17 Jan 03	10 May 04	481	2806.38	29 April 03	16 May 04	391	2294.50
22 Jan 03	n/a	n/a	n/a	30 April 03	n/a	n/a	n/a
22 Jan 03	n/a	n/a	n/a	30 April 03	24 May 04	404	2394.20
22 Jan 03	n/a	n/a	n/a	04 May 03	n/a	n/a	n/a
22 Jan 03	17 May 04	486	2848.85				
24 Feb 03	03 May 04	435	2527.17				
7 May 03	n/a	n/a	n/a				
[10 Jan 03]	[08 May 04]	[489]	[2843.09]	[23 April 03]	[18 May 04]	[397]	[2330.47]
SE = 6.04	SE = 1.25	SE = 4.93	SE = 28.13	SE = 1.35	SE = 1.41	SE = 1.46	SE = 10.78

undergo their molt to maturity, mate and extrude eggs is not as well defined as the timing of multiparous egg extrusion. In this study, pubescent females molted, mated and extruded their first egg clutch mid-December through mid-May with a mean extrusion date of 10 January 2003 (Table 3). Hilsinger (1976) and Donaldson (1977) report that pubescent females molt, mate and extrude eggs winter through spring. Similarly, pubescent females were found in mating embraces late December through June in southeastern Alaska (Stone, 1999). The longest time period for pubescent females to molt and mate is observed in Womans Bay, Kodiak, Alaska where pubescent females are found in mating embraces mid-November through mid-July (NMFS unpublished data, Fig. 1). The time period that pubescent females molt and mate may span up to 8 months, however peak molting and mating appears to occur in January and February as was observed in this study and by other researchers (Adams and Paul, 1983; Paul and Adams, 1984; Stone, 1999).

A comparison of egg extrusion timing of primiparous and multiparous crabs is often not made, but when examined among high latitude crabs, primiparous females extrude eggs earlier in the reproductive season than multiparous females. In this study, primiparous Tanner crabs extruded eggs on average 103 days earlier than multiparous crabs (Table 3). Primiparous snow crabs, a congener species to Tanner crabs, extrude eggs February through April and multiparous females extrude eggs later in the season, April through June (Sainte-Marie, 1993). Earlier egg extrusion by primiparous females is not limited to brachyuran species,

but also observed among anomuran species. For example, primiparous red king crabs *Paralithodes camtschaticus* (Tilesius, 1815) extrude eggs on average 2 months prior to multiparous females (Stevens and Swiney, 2007).

Crustacean eggs increase in size during development especially in the latter stages and increases in crustacean egg sizes have been attributed to increased water content throughout development (Davis, 1968; Wear, 1974; Davis, 1981; Sibert et al., 2004). Crab egg sizes do not increase steadily throughout development, but rather increases and

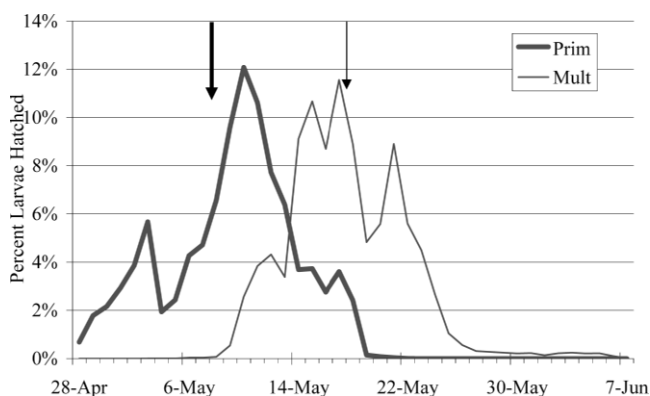


Fig. 5. Daily percent of primiparous and multiparous Tanner crab larvae hatched. Darker arrow on the left denotes the primiparous mean larval hatch date of 8 May 2004 and the lighter arrow on the right represents the mean multiparous larval hatch date of 18 May 2004. "Prim" denotes primiparous females and "Mult" denotes multiparous females.

decreases are observed, especially early in development (this study; Moriyasu and Lanteigne, 1998; Stevens 2006a; Stevens and Swiney, in review). The differences observed in egg areas before embryonic diapause in this study are statistically significant may not be biologically significant (Fig. 4). When differences were detected between the two groups the differences were small ranging from 0.006 to 0.009 mm. The differences observed may be an artifact of sampling considering the increases and decreases observed early in crab egg development. With such small differences in egg sizes it would not be possible to take a sample of eggs from a female and determine if the eggs come from primiparous or multiparous crabs. Alternately, empirical evidence and models suggest egg size can be used to estimate other life history characters and can be an indicator of egg energy content among some species (Jaekle, 1995). Giménez and Anger (2001, 2003) found that among *Chasmagnathus granulata* Dana, 1851 initial larval biomass depended in part on initial egg size and that both of these factors affected larval performance. In this study, multiparous Tanner crab eggs are initially larger than those of primiparous females (Fig. 4) and it is not yet known if the differences observed in initial egg sizes are related to differing energy content and larval viability.

Studies that compare reproductive cycles of primiparous and multiparous crabs consistently report that primiparous females have a longer incubation period than multiparous females. In this study, primiparous Tanner crabs exhibit an average incubation period of 489 days (approximately 16 months) and multiparous females exhibit an average incubation period of 397 days (approximately 13 months) (Table 3). Likewise, Donaldson and Adams (1989) report that primiparous Tanner crabs have an approximately 15-17 month incubation period and multiparous females have a 12 month incubation period, a month shorter than what was observed in this study. In the laboratory, primiparous snow crabs have an incubation period of 410 days whereas multiparous females had a shorter incubation period of 365 days (Moriyasu and Lanteigne, 1998). Similar to Tanner and snow crabs, red king crab primiparous females have a longer incubation period, 365 days, in comparison to 328 day incubation period observed for multiparous females. This trend is not surprising since primiparous crabs typically extrude eggs earlier in the reproductive season than multiparous females (see discussion above). In contrast to Tanner, snow and red king crabs, larger female Dungeness crabs (*Cancer magister* Dana, 1852) (> 140 mm CW) in Alaska may extrude eggs earlier in the season in comparison to smaller crabs (< 140 mm CW) (Swiney and Shirley, 2001), but direct comparisons are not possible since distinctions were not made between primiparous and multiparous Dungeness crabs.

Accumulated developmental degree-days vary among northern latitude crab species and in general, Tanner crabs appear to have higher developmental degree-day requirements. Among Tanner, snow and red king crabs primiparous females have more accumulated developmental degree-days than multiparous females. In this study, primiparous Tanner crabs accumulated 2843 developmental degree-days and multiparous females accumulated 2330 developmental

degree days. In aquaria at 1.75-1.85°C, primiparous snow crabs accumulated 716 developmental degree-days and multiparous females accumulated 674 developmental degree-days (Moriyasu and Lanteigne, 1998). In ambient waters, primiparous red king crabs accumulated 2601 developmental degree-days and multiparous red king crab accumulated 2482 developmental degree-days (Stevens and Swiney, 2007). Multiparous golden king crabs (*Lithodes aequispinus* Benedict, 1895) held in ambient water in Alaska accumulated 2269 developmental degree-days (Paul and Paul, 2001). Increasing temperature results in increasing accumulated developmental degree-days as was observed among blue king crabs reared at 2°C, 4°C, and 6°C with 1015, 1770 and 2503 accumulated developmental degree-days respectively (Stevens and Swiney, in review).

This is the first study to clearly illustrate an embryonic developmental diapause for Tanner crabs and to compare embryo development between primiparous and multiparous females. Crustacean diapauses either are an internal adaptation that synchronizes an organism's life cycle or are a defensive adaptation that permits the organism to endure conditions that would be oppressive or lethal for an active phase of the organism's life cycle (Alekseev and Starobogatov, 1996). The Malacostraca, which includes decapods, have very few species that exhibit diapause phases (Hairston and Cáceres, 1996). When diapauses are found among decapods, they appear to synchronize the organism's life cycle and may have evolved in some species to enhance larval survival in response to food availability in the plankton (Wear, 1974). Among crab species with a diapause phase, it is observed at the gastrula stage and occasionally a second diapause is observed later in development (this study; Wear, 1974; Lovrich and Vinuesa, 1993; Petersen and Anger, 1997; Moriyasu and Lanteigne, 1998; Comeau et al., 1999). In this study, embryos of primiparous Tanner crabs remained in embryonic developmental diapause for approximately 6 months in contrast to 3 months for the multiparous group (Fig. 3). Embryonic developmental diapauses in Tanner crabs appear to be a mechanism to synchronize embryo development and eclosion among primiparous and multiparous females.

Water temperature has been shown to be an important factor in embryo development and incubation period of crab species and only a slight change in temperature can alter incubation periods (Wear, 1974). For example, among red king crabs (Shirley et al., 1990), Dungeness crabs (Shirley et al., 1987), blue king crabs (*Paralithodes platypus* Brandt, 1850) (Stevens and Swiney, in review), and snow crabs (Mallet et al., 1993; Moriyasu and Lanteigne, 1998; Webb et al., 2007) incubation period decreases with increasing temperature. In addition, water temperature can have an effect on the presence or absence of embryonic developmental diapauses and this relationship can help explain differences in incubation periods observed among crabs reared in different temperatures. Snow crab females in colder water from the wild have an approximately 2 year incubation period with 2 embryonic developmental diapause phases (6 months minimum at the gastrula stage and 3-4 months at the eye-pigment formation stage) and crabs reared in the laboratory in warmer water temperatures have a 1 year

incubation period and no embryonic developmental diapause phases (Mallet et al., 1993; Moriyasu and Lanteigne, 1998). Blue king crabs reared at 2°C did not have an embryonic developmental diapause phase, however crabs reared at 4° and 6°C exhibited a diapause phase (Stevens and Swiney, in review). The relationship between water temperature and embryonic developmental diapauses is a complicated one. As is seen with snow and king crabs, water temperature can affect the presence or absence of diapauses however, Wear (1974) found that the length of diapause could not be shortened significantly by raising water temperatures and the response to increasing temperature followed that of other species only after the diapause period ended.

The duration of eclosion is innate, heritable and not easily perturbed by most environmental influences including temperature (Morgan, 1995) which has been illustrated for both blue king crab (Stevens, 2006b) and snow crabs (Webb et al., 2007) reared at different temperatures. The duration of eclosion varies by species and among high latitude crabs, brachyurans and anomurans exhibit different strategies. On average, eclosion duration for primiparous Tanner crabs is 7 days, 12 days for multiparous Tanner crabs, 11 days for snow crabs (Webb et al., 2007) and 11 days for Dungeness crabs in Alaska (author's observation). Alternately, lithodid crabs have extended eclosion durations; 29 days for blue king crabs (Stevens, 2006b), 31 days for red king crabs (Swiney and Stevens, in review) and 34 days for golden king crabs (Paul and Paul, 2001). Stevens (2006b) suggested that extended hatching observed among lithodid species may be evolutionary responses to conditions that were highly variable and unpredictable where as brachyurans respond to environmental cues and hatch over a shorter period of time which may be an adaptation to optimize larval transport.

Two differing triggers for *Chionoecetes* sp. eclosion have been proposed. Starr et al. (1994) suggest that eclosion among snow crabs lags two to three weeks behind the sedimentation peak of phytoplankton particles that originate from surface spring blooms. This relationship between phyto-detritus and eclosion could be an adaptive mechanism for these species to harvest the peak abundance of secondary production in high-latitude areas where plankton blooms show a high level of seasonality. Alternately, Stevens (2003) proposed that Tanner crab eclosion is controlled by an endogenous biological clock and coincides with the period of greatest tidal exchange to optimize larval transport rather than planktonic food availability.

This study was unique because to date, no one had compared mean eclosion dates between primiparous and multiparous Tanner crabs. Previous eclosion studies either did not distinguish between the reproductive states or only looked at multiparous females. Eclosion in this study was relatively synchronous with mean hatch dates of 8 May 2004 and 18 May 2004 for primiparous and multiparous females respectively (Fig. 5). This timing of eclosion is consistent with eclosion timing reported for Tanner crabs by other researchers (Hilsinger, 1976; Donaldson, 1977; Inceze et al., 1982; Donaldson and Adams, 1989; Stevens, 2003). Similarly, red king crab larval eclosion is relatively

synchronous, with eclosion of primiparous red king crabs in the laboratory occurring approximately 12 to 16 days earlier than multiparous females (Stevens and Swiney, in review). Similarly, in the field, bimodal hatching peaks of red king crab larvae in southeastern Alaska was observed and attributed to primiparous females hatching larvae earlier than multiparous females (Shirley and Shirley, 1989). The differences observed between mean eclosion timing of primiparous and multiparous females may result in one group having an advantage over the other due to planktonic food availability.

Botsford (1991) concluded that egg production and its impact on recruitment are critical aspects in crustacean fishery management, that our understanding is limited and that one way to improve management would be to use age dependent information. Results of this study clearly illustrate that primiparous and multiparous females have different reproductive cycles. The longer incubation period of primiparous females is the result of pubescent females undergoing their terminal molt and then mating mid-November through mid-July (NMFS unpublished data) whereas multiparous females are still brooding their egg clutches during this time and therefore eclosion must first occur after which they extrude a new clutch late April through May (NMFS unpublished data) (Fig. 1). Tanner crabs exhibit embryonic developmental diapauses of varying length depending on reproductive state which results in relatively synchronous eclosion. Reproductive potential of primiparous and multiparous crabs may differ due to the differences in their reproductive cycles. These differences need to be further studied and knowledge obtained utilized in models for effective stock assessment, fishery management plans and re-building plans for crab species.

ACKNOWLEDGEMENTS

I thank the numerous interns who maintained the crabs during this study. I gratefully acknowledge E. Wilson for her assistance in the laboratory. A special thanks goes to divers J. E. Munk and P. Cumiskey and skipper G. Edwards of the F/V *Big Valley* for crab collections, Dr. P. van Tamen and D. Barnard for statistical advice and C. Armistead who made the map. Drs. B. Stevens, T. Shirley and three anonymous reviewers provided critical reviews that improved this manuscript.

REFERENCES

- Adams, A. E., and A. J. Paul. 1983. Male parent size, sperm storage and egg production in the crab *Chionoecetes bairdi* (Decapoda, Majidae). *International Journal of Invertebrate Reproduction* 6: 181-187.
- Alaska Department of Fish and Game. 2004. Annual Management Report for the Commercial and Subsistence Shellfish Fisheries of the Aleutian Islands, Bering Sea and the Westward Region's Shellfish Observer Program, 2003. Regional Informational Report 4K04-03. 235 p.
- Alekseev, V. R., and Y. I. Starobogatov. 1996. Types of diapause in Crustacea: definitions, distribution, evolution. *Hydrobiologia* 320: 15-26.
- Botsford, L. W. 1991. Crustacean egg production and fisheries management, pp. 379-394. In, A. Wenner and A. Kuris (eds.), *Crustacean egg production*. A. A. Balkema, Rotterdam.
- Comeau, M., M. Starr, G. Y. Conan, G. Robichaud, and J. C. Therriault. 1999. Fecundity and duration of egg incubation for multiparous female snow crabs (*Chionoecetes opilio*) in the fjord of Bonne Bay, Newfoundland. *Canadian Journal of Aquatic and Fisheries Science* 56: 1088-1095.
- Davis, C. C. 1968. Mechanisms of hatching in aquatic invertebrate eggs. *Oceanography and Marine Biology Annual Review* 6: 325-376.

- . 1981. Mechanisms of hatching in aquatic invertebrate eggs II. Oceanography and Marine Biology Annual Review 19: 95-123.
- Donaldson, W. E. 1977. Kodiak Alaska Tanner crab (*Chionoecetes bairdi*) research. Commercial Fisheries Research and Development Act Project No. 5-35-R. 140 p.
- , and A. E. Adams. 1989. Ethogram of behavior with emphasis on mating for the Tanner crab *Chionoecetes bairdi* Rathbun. Journal of Crustacean Biology 9: 37-53.
- Giménez, L., and K. Anger. 2001. Relationships among salinity, egg size, embryonic development, and larval biomass in the estuarine crab *Chasmagnathus granulata* Dana, 1851. Journal of Experimental Marine Biology and Ecology 260: 241-257.
- , and ———. 2003. Larval performance in an estuarine crab, *Chasmagnathus granulata*, is a consequence of both larval and embryonic experience. Marine Ecology Progress Series 249: 251-264.
- Hairston, N. G., and C. E. Cáceres. 1996. Distribution of crustacean diapause: micro- and macroevolutionary pattern and process. Hydrobiologia 320: 24-11.
- Hilsinger, J. R. 1976. Aspects of the reproductive biology of female snow crabs, *Chionoecetes bairdi*, from Prince William Sound and the adjacent Gulf of Alaska. Marine Science Communications 2: 201-225.
- Incze, L. S., D. A. Armstrong, and D. L. Wencker. 1982. Rates of development and growth of larvae of *Chionoecetes bairdi* and *C. opilio* in the southeastern Bering Sea, pp. 191-218. In, B. Melteff (ed.), Proceedings of the International Symposium on the Genus *Chionoecetes*. Lowell Wakefield Symposia Series. Alaska Sea Grant Report 82-10.
- Jaecle, W. B. 1995. Variation in the size, energy content, and biochemical composition of invertebrate eggs: Correlates to the mode of larval development, pp. 49-76. In, L. McEdward (ed.), Marine Invertebrate Larvae. CRC Press, Inc., Boca Raton.
- Lovrich, G. A., and J. H. Vinuesa. 1993. Reproductive biology of the false southern king crab (*Paralomis granulosa*, Lithodidae) in the Beagle Channel, Argentina. Fishery Bulletin 91: 664-675.
- Mallet, P., G. Y. Conan, and M. Moriyasu. 1993. Periodicity of spawning and duration of incubation time for *Chionoecetes opilio* in the Gulf of St. Lawrence. International Council for the Exploration of the Sea C.M. 1993:K:26. 19 p.
- Morgan, S. G. 1995. The timing of larval release, pp. 157-191. In, L. McEdward (ed.), Marine Invertebrate Larvae. CRC Press, Inc., Boca Raton.
- Moriyasu, M., and C. Lanteigne. 1998. Embryo development and reproductive cycle in the snow crab, *Chionoecetes opilio* (Crustacea: Majidae), in the southern Gulf of St. Lawrence, Canada. Canadian Journal of Zoology 76: 2040-2048.
- Paul, A. J. 1984. Mating frequency and viability of stored sperm in the Tanner crab *Chionoecetes bairdi* (Decapoda, Majidae). Journal of Crustacean Biology 4: 375-381.
- , and A. E. Adams. 1984. Breeding and fertile period for female *Chionoecetes bairdi* (Decapoda, Majidae). Journal of Crustacean Biology 4: 589-594.
- , and J. M. Paul. 1992. Second clutch viability of *Chionoecetes bairdi* Rathbun (Decapoda: Majidae) inseminated only at the maturity molt. Journal of Crustacean Biology 12: 438-441.
- , and ———. 2001. The reproductive cycle of golden king crab *Lithodes aequispinus* (Anomura : Lithodidae). Journal of Shellfish Research 20: 369-371.
- , ———, and W. E. Donaldson. 1996. Shell condition and breeding success in Tanner crabs. Journal of Crustacean Biology 15: 476-480.
- Petersen, S., and K. Anger. 1997. Chemical and physiological changes during the embryonic development of the spider crab, *Hysa araneus* L. (Decapoda: Majidae). Comparative Biochemistry and Physiology B-Biochemistry & Molecular Biology 117: 299-306.
- Rohlf, F. J., and R. R. Sokal. 1981. Statistical Tables. W. H. Freeman and Company, New York. 92 p.
- Rugolo, L. J., E. A. Chilton, C. E. Armistead, and J. A. Haaga. 2006. Report to industry on the 2006 Eastern Bering Sea crab survey. AFSC Processed Rep. 2006-17, 61 p. Alaska Fisheries Science Center, NOAA, National Marine Fisheries Service, Kodiak Fisheries Research Center, Kodiak, Alaska 99615-7400.
- Sagalkin, N. H. 2004. Fishery management plan for the commercial Tanner crab fishery in the Kodiak District of Registration Area J, 2005. Alaska Department of Fish and Game, Fisheries Management Report No. 04-13. 29 p.
- . 2006. Annual report for the shellfish fisheries of Kodiak, Chignik and Alaska Peninsula Areas, 2005. Alaska Department of Fish and Game, Fisheries Management Report No. 06-48, Anchorage. 72 p.
- Sainte-Marie, B. 1993. Reproductive cycle and fecundity of primiparous and multiparous female snow crab, *Chionoecetes opilio*, in the Northwest Gulf of St. Lawrence. Canadian Journal of Fisheries and Aquatic Sciences 50: 2147-2156.
- Shirley, S. M., and T. C. Shirley. 1989. Interannual variability in density, timing and survival of Alaskan red king crab *Paralithodes camtschatica* larvae. Marine Ecology Progress Series 54: 51-59.
- , ———, and S. D. Rice. 1987. Latitudinal variation in the Dungeness crab, *Cancer magister*, zoeal morphology explained by incubation temperature. Marine Biology 95: 371-376.
- Shirley, T. C., S. M. Shirley, and S. Korn. 1990. Incubation period, molting and growth of female red king crabs: effects of temperature, pp. 51-63. In, B. Melteff (ed.), Proceedings of the International Symposium on king and Tanner crabs. Lowell Wakefield Symposia Series. Alaska Sea Grant Report 90-04.
- Sibert, V., P. Ouellet, and J.-C. Brêthes. 2004. Changes in yolk proteins and lipid components and embryonic growth rates during lobster (*Homarus americanus*) egg development under a simulated seasonal temperature cycle. Marine Biology 14: 1075-1086.
- Sokal, R. R., and F. J. Rohlf. 1995. Biometry-The Principles and Practice of Statistics in Biological Research. W. H. Freeman and Company, New York. 886 p.
- Somerton, D. A., and W. S. Meyers. 1983. Fecundity differences between primiparous and multiparous female Alaskan Tanner crab (*Chionoecetes bairdi*). Journal of Crustacean Biology 3: 183-186.
- Starr, M., J.-C. Theriault, G. Y. Conan, M. Comeau, and G. Robichaud. 1994. Larval release in a sub-euphotic zone invertebrate triggered by sinking phytoplankton particles. Journal of Plankton Research 16: 1137-1147.
- Stevens, B. G. 2003. Timing of aggregation and larval release by Tanner crabs, *Chionoecetes bairdi*, in relation to tidal current patterns. Fisheries Research 65: 201-216.
- . 2006a. Embryo development and morphometry in the blue king crab *Paralithodes platypus* studied by using image and cluster analysis. Journal of Shellfish Research 25: 569-576.
- . 2006b. Timing and duration of larval hatching for blue king crab *Paralithodes platypus* Brandt, 1850 held in the laboratory. Journal of Crustacean Biology 26: 495-502.
- , and K. M. Swiney. 2007. Hatch timing, incubation period, and reproductive cycle for primiparous and multiparous red king crab *Paralithodes camtschaticus* (Tilesius, 1815). Journal of Crustacean Biology 27: 37-48.
- , and ———. in review. Thermal effects on embryo development and hatching for blue king crab *Paralithodes platypus* Brandt, 1850 held in the laboratory, and a method for predicting dates of hatching. Journal of Shellfish Research.
- Stone, R. P. 1999. Mass molting of Tanner crabs *Chionoecetes bairdi* in a southeast Alaska estuary. Alaska Fishery Research Bulletin 6: 19-28.
- Swiney, K. M., and T. C. Shirley. 2001. Gonad development of southeastern Alaskan Dungeness crab, *Cancer magister*, under laboratory conditions. Journal of Crustacean Biology 21: 897-904.
- SYSTAT. 2002. SYSTAT Version 10.2. SYSTAT Software, Inc. Richmond, California.
- Wear, R. G. 1974. Incubation in British decapod crustacea, and the effects of temperature on the rate and success of embryonic development. Journal of the Marine Biological Association of the United Kingdom 54: 745-762.
- Webb, J. B., G. L. Eckert, T. C. Shirley, and S. L. Tamone. 2007. Changes in embryonic development and hatching in *Chionoecetes opilio* (snow crab) with variation in incubation temperature. Biological Bulletin 213: 67-75.
- Wenner, A. M., and J. Dugan. 1991. Estimating egg production in multi-brooding populations. pp. 333-355. In, A. Wenner and A. Kuris (eds.), Crustacean Egg Production. Crustacean Issues 7. A. A. Balkema, Rotterdam.

RECEIVED: 27 March 2006.

ACCEPTED: 22 August 2007.